From Semantic Seed to Object Mask: Controlled Selection of Diffusion Attention for Training-Free Segmentation

Supplementary Material

1. Protocol

The full protocol, with every registration, result and review in the order they happened, is `docs/ATTENTION_SEGMENTATION_EXPERIMENT_PROTOCOL.md` (113 sections); the registry of experiment families is `docs/ATTENTION_FUSION_AND_TOKEN_IDEAS.md`. This section summarizes the rules that every reported number obeys.

Fixed conditions. Stable Diffusion 1.5 generates every image from its released prompt and seed, with 30 scheduler steps, classifier-free guidance 7.5, a float32 model and float16 storage of compact attention. "Calls 0–30" are the 31 recorded UNet call indices, inclusive. CA uses the conditional branch for query and key, a softmax over all 77 positions of the padded attribution prompt, and the mean over every tokenizer piece of the class name within each head. All maps are combined on a 64×64 grid. Generated images and attention tensors are discarded after scoring; per-image TP, FP and FN at every threshold of the grid are kept, so that folds, bootstrap intervals and later analyses need no regeneration.

Folds. `configs/voc_sim_folds.csv` assigns the 490 VOC-Sim images to five class-stratified folds (about five images per class per fold), with a fixed seed, committed before any run. In each outer fold, the method parameter and the mask threshold are chosen on the other four folds and scored on the held-out fold; the reference is treated the same way.

Decision rule (v1.2). A candidate is promoted if it improves training-fold mIoU by at least 0.010, or by 0.005–0.0099 with at least 12 of 20 classes improved and no class losing more than 0.05; it must win at least four of five fold votes and give a positive pooled OOF gain. Within 0.002, the simpler setting wins (the reference, then fewer added operations, then fewer free parameters). Each family is screened against its own reference; family winners then go to a combined confirmation.

Registry rules. R1: only VOC-Sim selects; COCO-Cap is scored once per frozen pipeline. R2: the frozen CA/SA pair is the reference input. R3: folds and thresholds as above, on the 17-point grid 0.30–0.70. R4–R5: the decision and tie rules above. R6: per-image records, commit and environment for every shard. R7: each package runs in one environment and compares only with its own reference. R8: grids are committed before the run; later changes are logged and labelled post hoc. R9: every package reports fold choices, OOF scores, paired bootstrap intervals, per-class changes and the number of configurations searched. R10: training-free, few-shot and external-segmenter tracks are reported separately. R11: no token is trained on an image of the fold it is scored on.

Deviations. Four departures from these rules affect reported numbers, and each is labelled where it appears. (i) The A3 CA weighting was retained by owner decision after the registered selector kept its reference (SG24-A); all later studies (SG25–SG29) build on it. (ii) SG13's candidate cells use a 0–1 threshold grid by 0.005, registered before the run; its controls keep the 17-point grid. (iii) SG22–SG24 shards ran on a mix of V100 and A100 nodes, against R7 (Sec. 5). (iv) SG23's raw promotion depends on the order of tied candidates, which the registration did not fix (Sec. 5).

2. Every registered family

Table 1 lists the 45 comparisons of Fig. 2 of the main paper with their reference, plotted cell, number of candidates and both changes. Every value is recomputed from the committed per-image shards of the named package (`scripts/paper/figure_decision_map.py`; values in `outputs/segmentation/paper_figures/decision_ma`). Three choices keep the comparison uniform. Each point compares two cells of one run, so no environment difference enters. Each cell is scored at its own full-VOC threshold on the 17-point grid, which places the point but is in-sample; the promotion decision is the out-of-fold rule of the protocol section given. Cells that repeat another package's operator by design (SG26 S1/S2, SG27 S1/S2, which reproduce SG25) are excluded from the best-variant choice.

What each family tested. *Graph restriction.* SG1 estimates, for each uncertain pixel, the probability that a walk reaches confident foreground before confident background. SG3 preserves pixels on which the call–head observations agree and propagates mainly elsewhere. SG9 replaces each affinity by a reciprocal form of A and $A^\top$ (geometric mean, minimum or soft gate). SG10 scores each

Family	Sec.	Reference	Plotted cell	Cand.	ΔVOC	ΔCOCO	Decision
(a) Operators on the propagated map							
P1 two passes	43	W/REF	W/P1	1	+1.20	+0.73	promoted
R2 DenseCRF	44	W/REF	W/REF+R2	1	+1.61	+1.77	promoted
SG1 absorbing walk	71	W/REF	W/SG1/f90_b10_r1	12	-4.30	-8.22	–
SG3 confidence lock	71	W/REF	W/SG3/f95_b05_r0_n1	24	-1.20	+0.21	–
SG5 contrastive seed	70	W/REF	WN/REF	5	+0.62	-2.66	–
SG8 observation consensus	78	W/REF	W/C/rho0p80	15	+0.81	+0.35	promoted
SG9 reciprocal links	78	W/REF	W/R1_geometric	3	+0.26	+0.09	–
SG10 bottleneck	79	W/REF	W/B3_G2_q95	16	-0.59	-0.22	–
SG2 consensus graph	85	W/REF	W/G_cv_g20_n1	12	+0.98	+0.61	promoted
SG4 SA regions	85	W/REF	W/SPEC32_rmean	18	+0.12	-0.66	–
SG11 robust consensus	89	W/REF	W/SG11/trim20/rho0p80	6	+0.98	+0.48	promoted
SG12 observation filtering	89	W/REF	W/SG12/top50	4	+0.99	+1.00	promoted
SG13 region inference	90	W/REF	W/SG13/R1	3	-2.03	-1.66	–
R1 (backlog)	83	REF/one_pass	R1/w100	3	+0.41	+1.22	–
M2 (backlog)	83	REF/one_pass	M2/k2	3	-20.66	-20.32	–
M1 (backlog)	83	REF/one_pass	M1/ca_prototype	1	-39.75	-34.48	–
G4 (backlog)	83	REF/one_pass	G4/s070	3	-41.59	-36.24	–
G1 (backlog)	83	REF/one_pass	G1/tau090	2	-9.25	-8.22	–
H1 (backlog)	83	REF/one_pass	H1/proportional	1	-0.75	+0.92	–
W4 (backlog)	83	REF/one_pass	W4/converged	1	-17.75	-25.08	–
W2 (backlog)	83	REF/one_pass	W2/k6	3	-0.40	+0.05	–
W1 (backlog)	83	REF/one_pass	W1/T050	3	+0.02	+0.03	–
BG1 (backlog)	83	REF/one_pass	BG1/g050	5	-5.49	-4.38	–
F2 (backlog)	83	REF/one_pass	F2/a075	4	-0.70	-0.37	–
P9 (backlog)	83	REF/one_pass	P9/s050_bg0	6	-1.05	+3.12	–
P5 (backlog)	83	REF/one_pass	P5/n2	4	+0.07	-1.19	–
P3 (backlog)	83	REF/one_pass	P3/t400	4	-0.16	-1.27	–
SG14 combined consensus (vs SG11)	92	W/SG11/trim20/rho0p80	W/SG14	1	+0.03	-0.02	–
SG15 refinement (vs SG11)	92	W/SG11/trim20/rho0p80	W/SG11/trim20/rho0p80/SG15-B	4	+0.05	-0.88	–
SG17 filtered 2nd pass (vs P1)	97	W/P1	W/SG17-Q70	3	-0.06	-0.52	–
SG18 middle calls	97	W/REF	W/T1	3	+0.73	+1.13	promoted
SG21 head weighting (vs T1)	101	SG18/T1	SG21-DIST	4	-0.19	+0.02	–
SG23 middle-call SG2 (vs MID one-pass)	102	MID-R0	MID-SG2	3	+0.61	+0.06	promoted[a]
SG28 PEARL-inspired (vs A3 P2)	112	P0	B2	11	+0.27	+0.55	–
(b) Complete pipelines with DenseCRF							
SG2 + CRF (vs FTF)	89	W/P1+R2	W/SG2+R2	1	-0.21	-1.10	–
SG12 + CRF (vs FTF)	92	W/P1+R2	W/SG12/top50+R2	1	-0.01	-0.56	–
T1 + CRF (vs FTF)	98	W/P1+R2	W/T1+R2	1	-0.13	+0.89	–
SG22 middle calls + CRF (vs FTF)	102	ALL-P2+R2	MID-P2+R2	3	+0.53	+0.67	promoted
SG23 middle-call SG2/11/12 + CRF	102	ALL-P2+R2	MID-SG11+R2	6	+0.45	+0.32	–
SG24-A CA weighting (A3)	103	A0/FINAL	A3/FINAL	4	+0.58	+1.83	owner[b]
SG24-B 16 CA modules	103	B0-A3/FINAL	B1-A3/FINAL	1	-0.93	+0.82	–
SG25 multi-resolution SA	104	FINE-P2+R2	32-to-64+R2	5	-0.14	+0.96	–
SG26 coarse-guided SA	105	S0+R2	S5+R2	4	-0.42	+2.00	–
SG28 PEARL-inspired + CRF	112	P0+R2	P3+R2	11	-0.08	-0.78	–
SG27 late fusion	109	S0+R2	S8+R2	39	+0.16	+1.22	–

Table 1. **Every registered family in Fig. 2.** Changes in mIoU points against the reference cell of the same run; VOC at each cell's full-VOC threshold (in-sample), COCO at that threshold (frozen transfer). The plotted cell is the promoted one, or else the best of the family's candidates (Cand.) on full VOC. Sec.: protocol section with the registration and result. [a]Promotion depends on the unregistered tie order (Sec. 5). [b]Not promoted by the registered selector; retained by owner decision.

pixel by its widest-path (bottleneck) connectivity to the confident seed. SG4 and SG13 partition the SA graph into regions and label whole regions from CA evidence. *Backlog* (Sec. 83): seeded progressive expansion (P9), a propagated background map (BG1), iteration to convergence (W4), multi-resolution fusion (H1), DiffSeg-style attention merging (G1), seeded region growing (G4), CA-weighted single and multiple prototypes (M1, M2), DenseCRF with SA edges (R1), a heat kernel over pass counts (P3), a symmetric normalized affinity (P5), a column-mass gate (F2), entropy-weighted observations (W1) and per-image head selection by agreement with the seed (W2). *Observation reliability.* SG2 weights each SA edge by its consistency across observations; SG8 combines the 248 propagated maps by per-pixel trimmed means, variance gating or sign agreement (promoted: sign agreement of a fraction $\rho = 0.8$); SG11 aggregates observations robustly (trimmed mean, median) where they agree; SG12 discards the least consistent observations. SG14 and SG15 combine or refine SG11. SG17 filters the second pass's additions by their reliability. *Time, heads and scale.* SG18 restricts SA to call windows; SG21 weights SA heads by structural roles; SG22 and SG23 cross the window with the number of passes and with the reliability operators, before and after DenseCRF. SG24 weights CA modules by their agreement with SA (A) or restores the 32-pixel CA modules (B). SG25–SG27 propagate through coarser SA modules alone, as guidance, or as late fusion. SG28 adds PEARL-inspired components (Procrustes align-

ment, boundary-aware Laplacians, cross-scale trust) [2].

3. Per-class effects

Figure 1 shows the per-class change of fourteen within-run comparisons, from the committed per-class tables (`scripts/narmyn/attention/cross_package_classes.py`). Three observations support the main text. First, the promoted raw-stage operators (top block) resemble the second pass on VOC-Sim: over the 20 classes, the change from SG2 correlates with the change from a second pass at 0.88 and that from SG12 at 0.82 (SG11: 0.44; SG8: 0.34). This is consistent with their acting through the same mechanism, object completion. Second, after DenseCRF (bottom block) the pattern weakens, and only the middle-call window and A3 keep a positive VOC-Sim mean. Third, a change's per-class effect on VOC-Sim predicts its effect on COCO-Cap poorly: the correlation over classes is between -0.01 and $+0.46$ for every comparison. Person is the most volatile class on VOC-Sim, with seven gains and seven losses beyond ± 0.005; potted plant and chair lose most often (eight comparisons each), and bicycle loses twice as often as it gains; on COCO-Cap, dining table loses in nine of fourteen comparisons. Figure 2 gives the class detail of the SG22 middle-call decision.

4. Exploratory record

Before the fold protocol existed, seventeen exploratory experiments (B1–B17) chose settings on the same images they reported. They motivated the controlled screens but are never used as evidence in the paper: CA time windows and resolutions (B1), single and leave-out CA layers (B2), joint time and resolution (B3), the exhaustive 65,535 CA layer subsets (B4), threshold and normalization (B5), heads, time–layer, fusion and cleanup (B6), SA screens and progressive SA (B7–B10), component ablation and COCO-Cap transfer of an earlier final method (B11–B13), CA call windows (B14), CA-only screens (B15), OVAM's released masks (B16), and OVAM's optimized token with DenseCRF (B17). Their maxima are optimistic by construction. The evidence package (`scripts/narmyn/attention/build_evidence.py`) lists each with its source files and the controlled screen that later tested the same idea.

5. Reproducibility and audits

Environments. Every shard records its commit and software environment. Packages ran in one of two conda environments, `gen` (PyTorch 2.4.1) and `trdp` (PyTorch 2.6 and NumPy 1.26 at the time of the runs), on V100 or A100 GPUs. Rule R7 requires one environment per package, and every comparison in the paper is within one package. Two exceptions are known: SG22–SG24 shards ran on a mix of V100 and A100 nodes, and the SG17, SG18 and SG21 shards do not record their environment. Regenerating images in another environment changes per-image counts slightly through floating-point differences in attention; the same two-pass-plus-DenseCRF pipeline, for example, scores 0.8132–0.8135 forced OOF across packages. We therefore never compare cells across packages.

Independent replication. Each package was reviewed by a second analyst, who re-derived its decisions from the per-image shards before the result entered the protocol; selected packages were also regenerated independently. The B2 token packages were independently regenerated (protocol Section 65), and the token optimizer was audited against OVAM's released code, with one recipe deviation recorded (Section 69). For SG17–SG27, every report was regenerated from the committed per-image shards: the regenerated tables match the committed ones byte for byte apart from line endings and last-digit floating-point differences, and every number quoted in the protocol matches. One dependence was found. SG23's raw promotion of MID-SG2 depends on the order in which tied candidates are visited, which the registration did not fix: the finalizer's order gives 5/5 fold votes, a neutral order 4/5, and the reverse order 3/5, in which case the reference is retained. SG23 is therefore marked as tie-order sensitive in Fig. 2 and Tab. 1. The same check found no effect of tie order on SG22.

The A3 affinity orientation. SG24's A3 score compares each CA module's query–key affinity with a resized SA affinity. In the implementation (`src/segmentation/sg24_ca.py`), the resized SA affinity is not transposed back, so the score is $\langle M^\top, P\rangle$ rather than GoCA's $\langle M, P\rangle$ [1]. We measured the effect on four images: the per-call module weights change by 0.9–2.0% (relative) and the call-averaged weights by at most 0.005. All A3 numbers in the paper are those of the implemented operator; a corrected replay of the A3 control is planned.

Sources of figures and tables. Fig. 1: `scripts/narmyn/attention/select_figure_images.py` (rule), `figure_maps.py` (regeneration; each final-mask IoU is checked against the SG5 shards) and `scripts/paper/figure_pipeline_stages.py`. Fig. 2 and Tab. 1: `scripts/paper/figure_decision_maps.py`. Propagation-depth figure: `scripts/narmyn/attention/figure_passes_maps.py` (rule and regeneration, checked against the SG22 shards) and `scripts/paper/figure_passes.py`. Per-class figure: `scripts/narmyn/attention/cross_package_classes.`

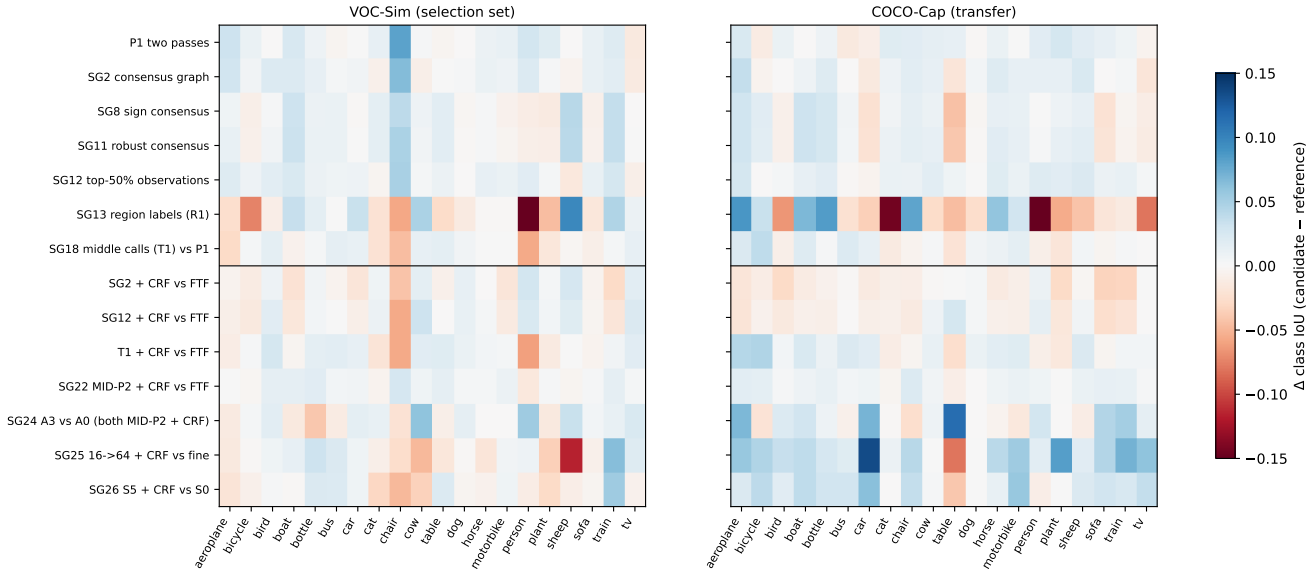


Figure 1. **Per-class IoU change, candidate minus same-run reference.** Rows: fourteen comparisons, raw operators above the line and complete pipelines with DenseCRF below; VOC-Sim at each cell's full-VOC threshold, COCO-Cap at that threshold (frozen). Blue: gain; red: loss; colour saturates at ± 0.15.

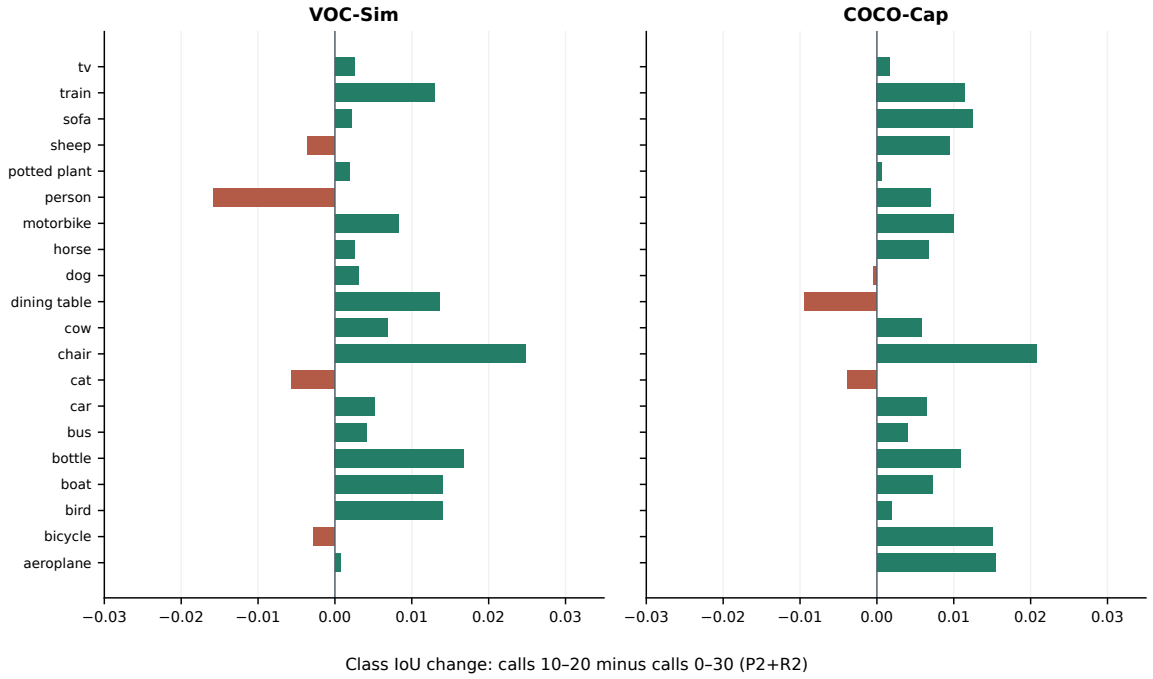


Figure 2. **The middle-call gain is heterogeneous.** Per-class IoU change for MID-P2+R2 versus ALL-P2+R2 in the same SG22 evaluation, with the threshold chosen on VOC only. Chair improves on both datasets; VOC person loses 0.016.

The remaining figures: `scripts/paper/build_story_figs.py`. No figure script selects a method or changes a result.

6. Additional qualitative figures

Figure 3 extends the failure columns of Fig. 1 by a fixed rule: the four classes with the lowest class IoU of the training-free pipeline on each dataset, each shown by its

median-IoU image. No image was inspected before the choice. Two of the eight (VOC dining table and COCO person) also appear in Fig. 1. The COCO-Cap columns show three different failures. For the chair, propagation leaks into the wooden wall behind it. For the dining table, which fills the frame, the mask covers the plate and bread on it but not its surface. For the bottle, the mask covers both bottles, but only one is annotated, so part of the error is in the label rather than the mask.

CA resolution. Figure 4 illustrates the CA resolution decision of Sec. 5 of the main paper on one image.

Propagation depth. Figure 5 applies both selection rules of the main paper's propagation-depth figure to both datasets.

References

[1] Benyuan Meng, Qianqian Xu, Zitai Wang, Xiaochun Cao, Longtao Huang, and Qingming Huang. Making training-free diffusion segmentors scale with the generative power. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 35861–35871, 2026.

[2] Gensheng Pei, Xiruo Jiang, Xinhao Cai, Tao Chen, Yazhou Yao, and Byeungwoo Jeon. PEARL: Geometry aligns semantics for training-free open-vocabulary semantic segmentation. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 17927–17937, 2026.

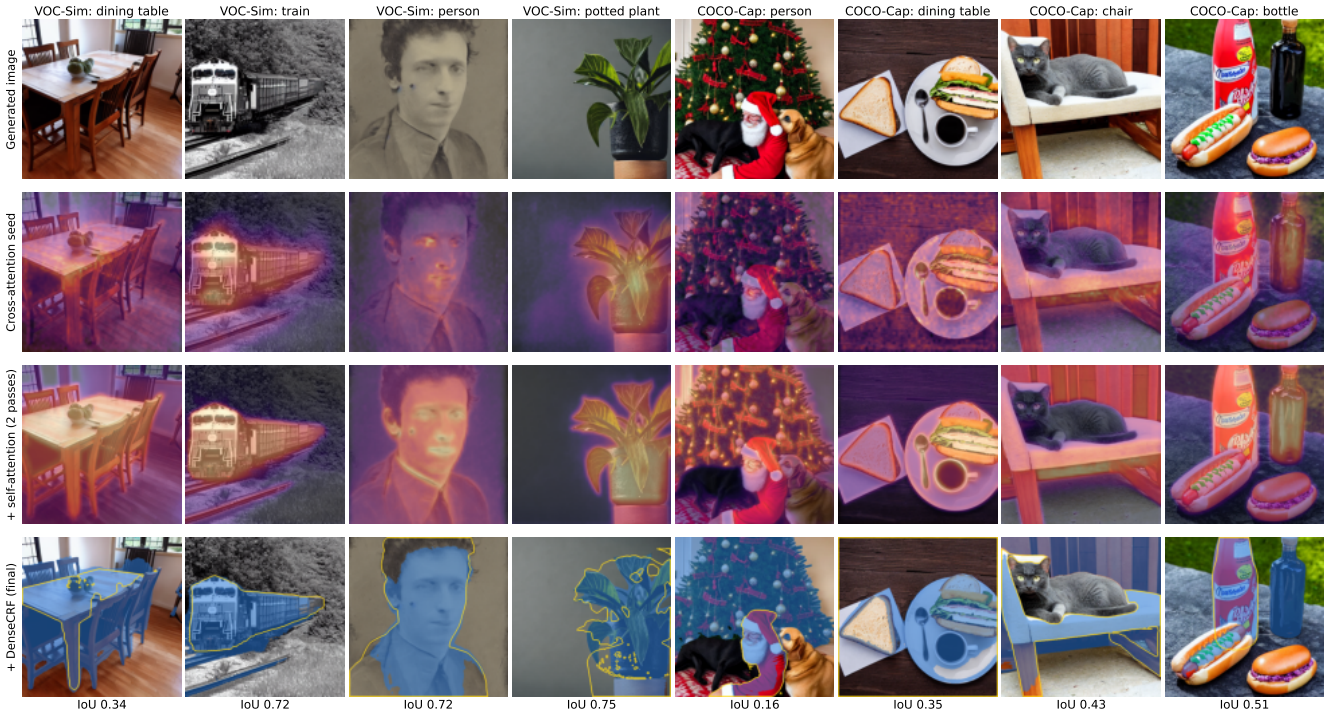


Figure 3. **The weakest classes, one median image each.** Rows as in Fig. 1: generated image; CA seed; two SA passes; final mask after DenseCRF at 0.45 (blue) with the ground-truth outline (yellow) and the image IoU. Columns: the four lowest class IoUs on VOC-Sim, then on COCO-Cap, from the lowest.

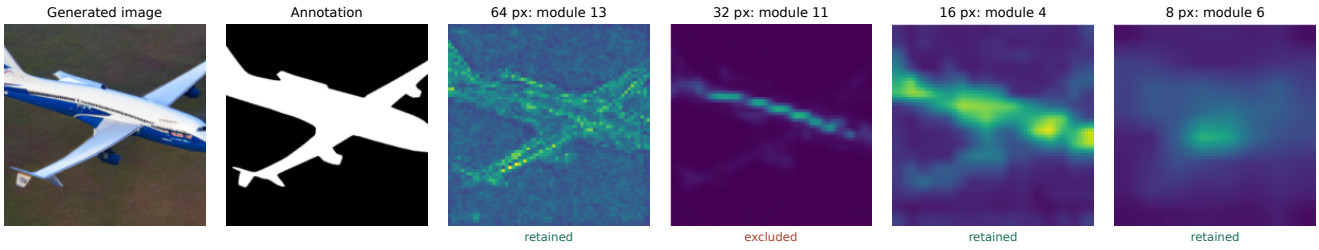


Figure 4. **Why CA resolution is a real design choice.** One VOC-Sim airplane generated image and annotation (left), followed by representative head-averaged CA modules from the same image and all calls. All heatmaps use one color scale. The selected seed retains 8, 16 and 64-pixel modules and omits the 32-pixel group. This visual example motivates the controlled resolution test; the fold-level evidence is in Sec. 5 of the main paper.

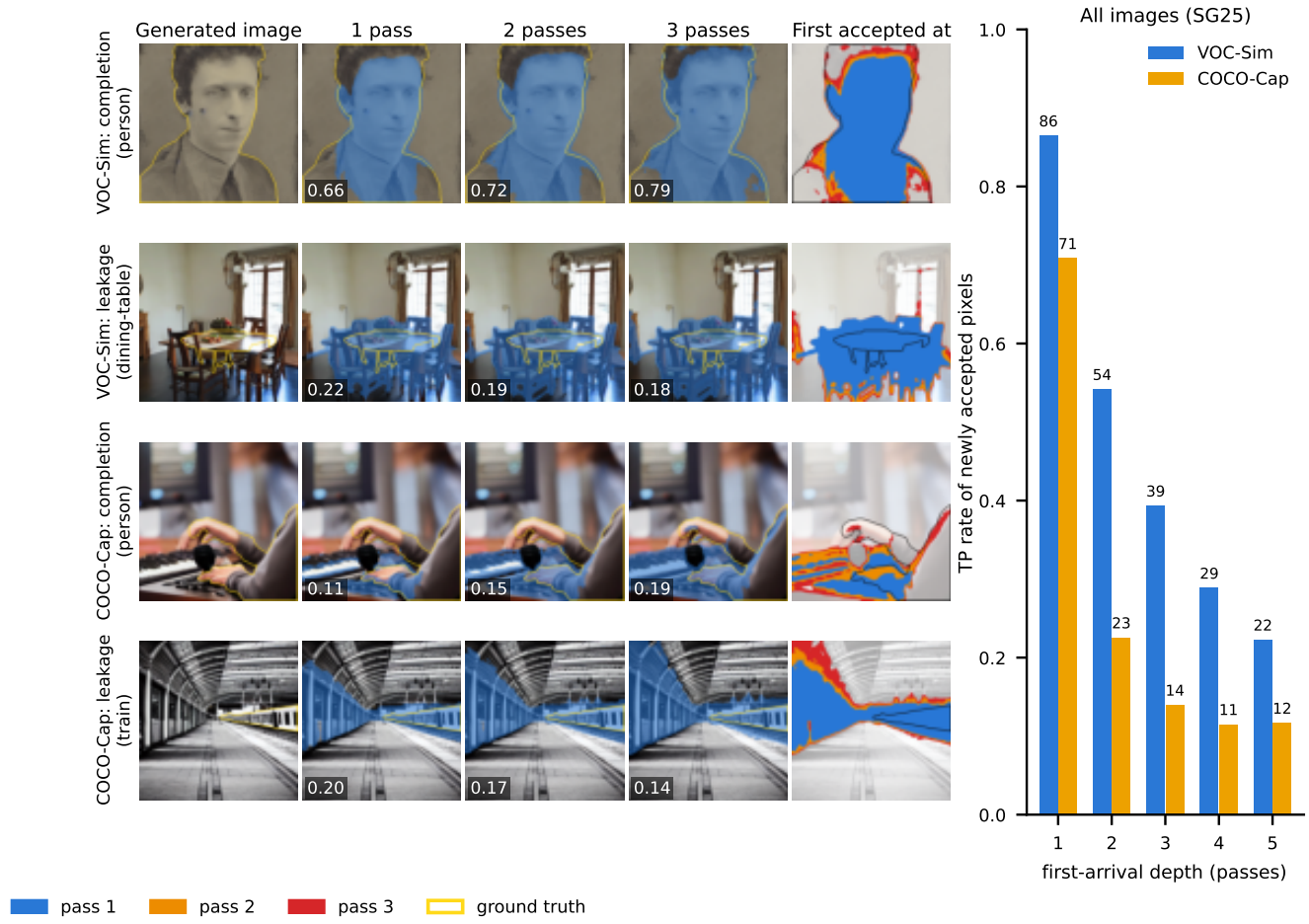


Figure 5. **One, two and three SA passes, both rules on both datasets.** Rows: on each dataset, the image at the 90th percentile of the two-pass gain (completion) and at the 10th percentile of the three-pass gain (leakage), from the all-call raw maps of SG22 at threshold 0.45. Columns and right panel as in the main paper.